Project Documentation

C64 Keyboard Connector PCB

Project number: 224

Revision: 0

Date: 04.06.2026



C64 Keyboard Connector PCB Rev. 0

Module Description

Introduction

The purpose of this PCB is to connect the C64 Keyboard (Project. No. 223 or 225) to the C64/Ultimate64 Keyboard Connector.



Figure 1: C64 Keyboard Connector PCB with ribbon cable

The connector for the ribbon cable is a MicroMatch series connector (red). They are low profile, so the cable does not collide with the side of the case when connected to a C64 long board. The C64 Keyboard has a 2x10p pin header, so the ribbon cable has two different types of connectors attached.

Dimensions

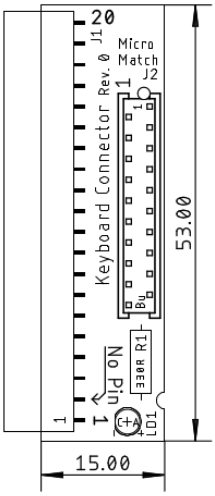


Figure 2: Dimensions of the PCB

Pinouts

J1 – Connector to mainboard

2.54mm pitch socket, 90°

Pin	Signal
1	GND
2	No pin
3	/RESTORE
4	+5V
5	ROW3
6	ROW6
7	ROW5
8	ROW4
9	ROW7
10	ROW2
11	ROW1
12	ROW0
13	COL0
14	COL6
15	COL5
16	COL4
17	COL3
18	COL2
19	COL1
20	COL7

J2 – Ribbon Cable Connector

MicroMatch Socket, 20 pins

Pin	Signal
1	GND
2	No pin
3	/RESTORE
4	+5V
5	ROW3
6	ROW6
7	ROW5
8	ROW4
9	ROW7
10	ROW2
11	ROW1
12	ROW0
13	COL0
14	COL6
15	COL5
16	COL4
17	COL3
18	COL2
19	COL1
20	COL7

Cable

The cable is a 20p ribbon cable of 45cm with one side attached to a 2x10p IDC socket with strain relief on the keyboard side and a 20 pin MicroMatch connector (male) on the side of this PCB. The cable is shown in the /Cables/pdf/ sub-folder of this repository, drawing number 233-3-01-00 (KB_Cab.pdf).

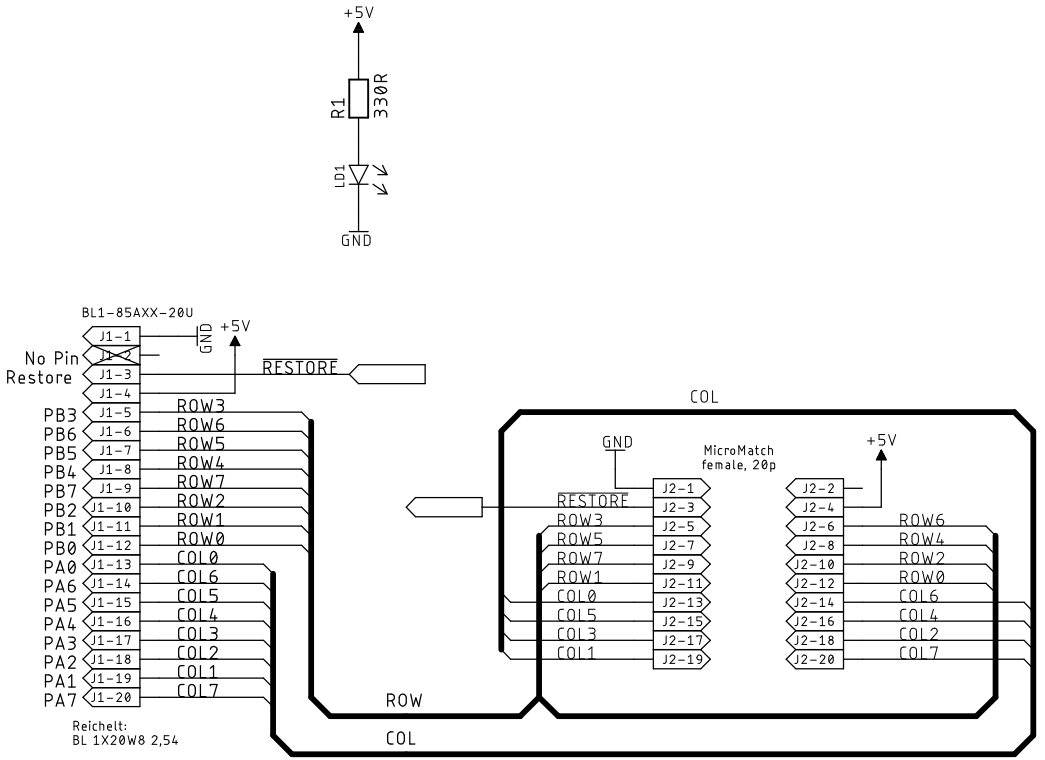
Both connector types are easy to attach, since they are both IDC (insulation-displacement contact), so you only require a vice to compress them, after the ribbon cable is positioned.

The length of 45cm is the longest possible, calculated for a C64 longboard (the keyboard connector is top left on the mainboard). For short boards, 30cm is long enough

Revision History

Rev. 0

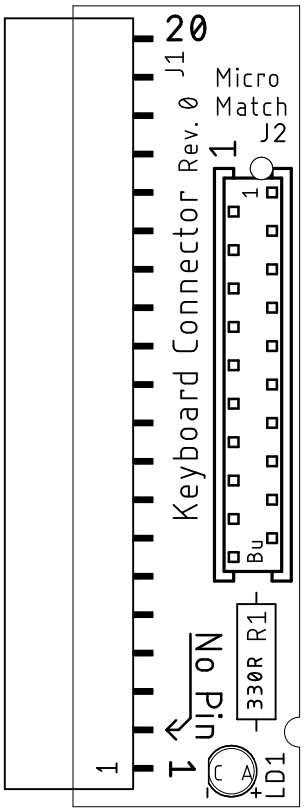
- Working prototype



open source hardware

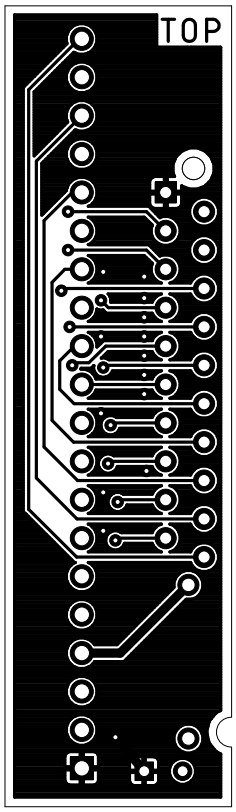
Title:	C64 Keyboard Connector PCB	Doc.-No.:	224-1-01-00
		Draft.:	Sven Petersen
Date:	20.02.2025 11:05	Rev.: 0	Page 1/1
File:	Keyboard_C64_connector_PCB		
	http://github.com/svenpetersen1965		A4

Sven Petersen 2025	Doc.-No.: 224-2-01-00	
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Keyboard_C64_connector_PCB		
20.02.2025 19:23		Rev.: 0
placement component side		

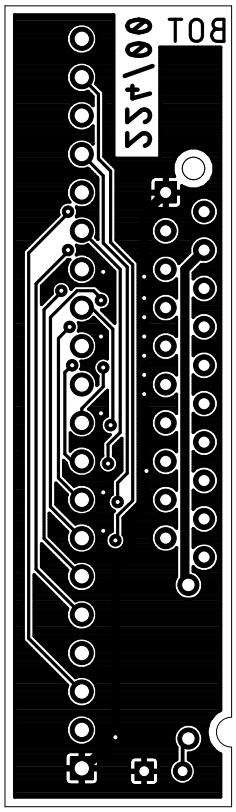


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20.02.2025 19:23		Rev.: 0
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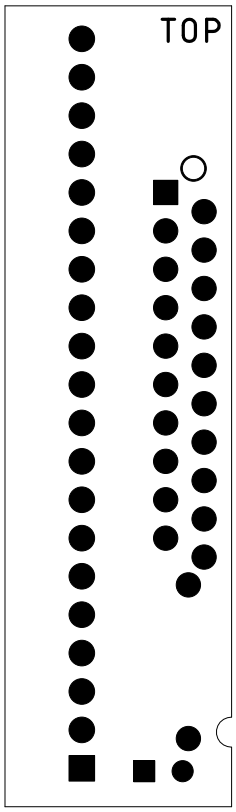
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20.02.2025 19:23		Rev.: 0
top	[redacted]	



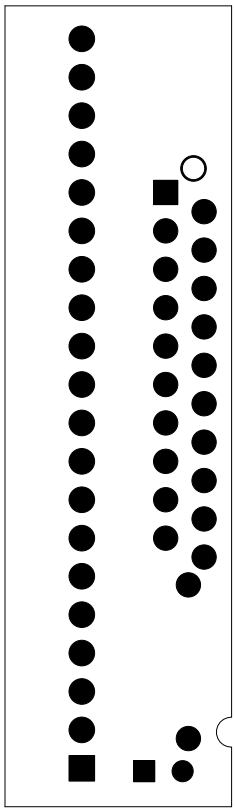
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Keyboard_C64_connector_PCB		
20.02.2025 19:23		Rev.: 0
bottom	[redacted]	



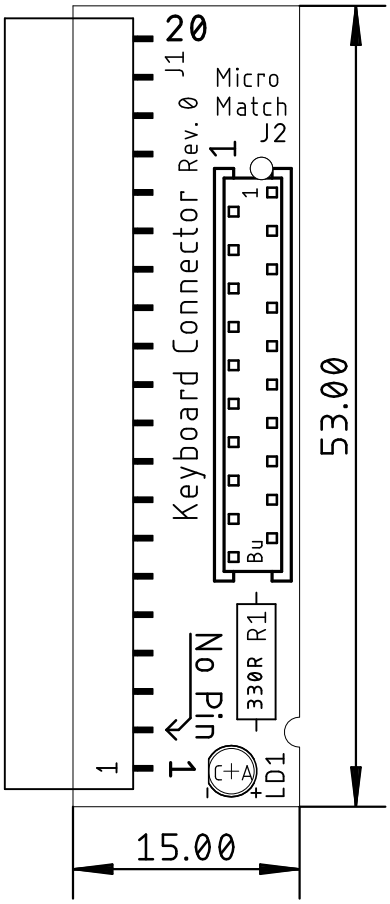
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Keyboard_C64_connector_PCB		
20.02.2025 19:23		Rev.: 0
stopmask component side		

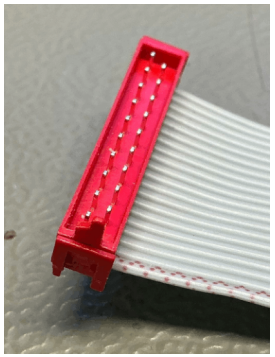


Sven Petersen 2025	Doc.-No.: 224-2-01-00	
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Keyboard_C64_connector_PCB		
20.02.2025 19:23		Rev.: 0
stopmask solder side		

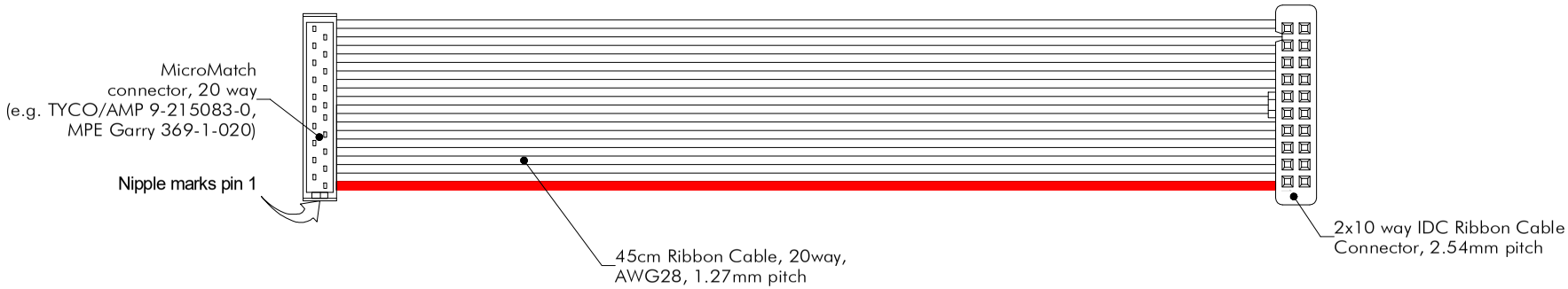


Sven Petersen 2025	Doc.-No.: 224-2-01-00	
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Keyboard_C64_connector_PCB		
20.02.2025 19:23		Rev.: 0
placement component side measures		





Detail MicroMatch coonector



This is the length for a long board C64 mainboard.
It varies for a short board or an Ultimate64.

open source hardware	Titel: C64 Keyboard Ribbon Cable	Bearb.: Sven Petersen
		Datei: KB_Cab.SKF
	Dok.-Nr.: 224-3-01-00	Rev.: 0
	Datum: 05.06.2026 14:33	M 1 mm = 1 mm
	https://github.com/svenpetersen1965	

C64 Keyboard Connector PCB Rev. 0
Bill of Material Rev. 0.0

Pos.	Qty	Value	Footprint	Ref.-No.	Comment
1	1	224-2-01-00	2 Layer	PCB Rev. 0	2 layer, Cu 35µ, HASL, 53.0mm x 15.0mm, 1.6mm FR4
	1	LED, green, 3mm	3MM	LD1	Standard LED, 3mm, green
	1	330R	R-10	R1	Metal film resistor, 1/4W & 10% or better
	1	BL1-85AXX-20U	BL1-85AXX-20U	J1	1x20 pin socket connector, 90°, e.g. MPE Garry, Reichelt BL 1x20W8 2,54
	1	369-1-020 or 2-215079-0	MICMA20B	J2	20 pin MicroMatch, femal, PCB-mount, e.g. MPE Garry, Reichelt MPE 369-1-020 or TE Connectivity 2-215079-0
	1	45cm ribbon cable		cable	20 way, AWG28, 1.27mm
	1	IDC header Micro Match, 20-pin		cable	Reichelt MPE 371-1-020 or TE Connectivity 9-215083-0
	1	20-pin IDC socket with strain relief, 2.54mm		cable	e.g. Reichelt PFL 20